

Graduation Engineering Project · MEF (GDDE) & ITC

Daily Price & Inflation Tracking for Cambodia

A simple, automated system to track daily cost-of-living changes in Cambodia and solve the 1–2 month waiting delay of traditional surveys.

PRESENTER

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What is CPI (Consumer Price Index)?

In Simple Words: The Cost of Living

- **What it is:** CPI measures how much prices change over time for everyday goods and services that families buy.
- **The Shopping Basket:** It tracks the price of a typical family basket: rice, pork, vegetables, cooking oil, gasoline, housing, and medicine.
- **What is Inflation?** When this total basket becomes more expensive compared to last month or last year, that price rise is called **inflation**.

Why Does CPI Matter for Cambodia?

- **Family Wellbeing:** It shows directly whether life is becoming harder or more affordable for everyday citizens.
- **Government Policy:** The Ministry of Economy and Finance (MEF) uses CPI to adjust minimum wages, plan social support, and manage the national budget.
- **National Bank of Cambodia (NBC):** Uses CPI to protect the purchasing power and stability of the Khmer Riel.

Problem Statement: Why the Current Way is Too Slow

The Problem

⚠️ The 1–2 Month Delay

- **Manual Data Collection:** Traditional government CPI relies on survey workers physically visiting markets once a month with clipboards.
- **Slow Publishing:** Compiling these paper surveys takes **30 to 60 days** after the month ends before results are published.
- **Blind to Sudden Jumps:** If food or gasoline prices spike sharply today, leaders do not see the official numbers until 1 or 2 months later.

⚠️ High Risk for Cambodian Families

- **Over 70% Spent on Food & Fuel:** Most Cambodian families spend the majority of their income on food and transportation, which fluctuate daily.
- **Dual Currency Impact:** Store prices switch between US Dollars and Khmer Riel; daily currency changes affect real costs immediately.
- **Delayed Help:** Waiting 60 days for data means government assistance programs arrive too late to protect vulnerable households.

Project Objectives: What We Want to Achieve

1. Daily Automatic Collection

Replace slow manual surveys by automatically gathering thousands of prices from online stores, markets, and official feeds every single morning.

2. Real-Time Price Index

Calculate national cost-of-living changes daily so government leaders can see inflation trends in real time without waiting 60 days.

3. Early Month-End Forecast

Provide an early-warning projection that estimates this month's inflation while the month is still ongoing, helping leaders act early.

The Proposed Solution: How It Works

Our Solution

Step 1: Automatic Collection & Cleaning

- **Collects 20+ Sources Daily:** Checks major supermarkets, pharmacies, fuel stations, bus travel, and the official MEF exchange rate.
- **Cleans & Standardizes:** Converts all US Dollar prices into Khmer Riel and calculates prices per standard unit (per 1 kg, per 1 liter).
- **Smart Product Matching:** Makes sure identical products from different stores match, and small packages aren't confused with large ones.

Step 2: Daily Inflation & Early Forecast

- **Daily Inflation Calculation:** Computes fair daily price changes across all 12 official government consumer categories.
- **Early Forecast:** Uses fast-moving food and fuel price trends to predict month-end inflation early with high accuracy.
- **Visual Dashboards:** Displays live, easy-to-read charts and reports for ministry decision-makers.

What I Am Doing Right Now (Project Progress)

Current Work

1. Running the Automated System Every Morning

The system is live and automatically collects over 31,000 prices every morning at 2:00 AM.

✓ 100% Live & Working

2. Cleaning, Currency Conversion & Product Categorization

Every product is cleaned, converted to Khmer Riel, and sorted into official categories daily.

✓ 100% Operational

3. Calculating Daily Inflation & Early Month Projections

Daily price indices and month-end early forecasts are generated automatically every day.

✓ 100% Implemented

4. Monitoring Trends on Live Visual Dashboards

Visual charts and summary tables are available for supervisors to review price movements.

✓ 100% Active

5. Preparing Graduation Thesis & Defense

All 5 report chapters are written; currently doing final review for defense on September 30, 2026.

✎ Final Polish